

Tilak Nanavati

Based in Ahmedabad, Gujarat, India
LinkedIn: tilak-nanavati

Mobile: +91 95868 32540
Email: tilak.nanavati@gmail.com

PROFILE

Gen AI Tech Lead & Principal Software Engineer with **7+ years** building and shipping production-grade AI systems across enterprise-scale platforms. Currently leading GenAI architecture at **Cognizant**, designing **agentic AI frameworks** (AWS Bedrock/AgentCore, Claude Sonnet) and **RAG pipelines** that automate complex data-migration workflows. Previously at **CleverDev/OpsHub** as Principal Engineer for **5 years**, where delivered **12+ AI-assisted analytical features**, drove **50% system efficiency gains** on 10M+ record datasets, and built LLM-powered platforms serving **100K+ queries/day** across 5+ business units. Deep hands-on expertise across the full AI lifecycle — **multi-agent orchestration**, **LangChain / LangGraph**, **vector databases**, **MCP servers**, and **SQL-layer performance engineering**. MTech Gold Medalist, DA-IICT (CPI 9.60).

SKILLS SUMMARY

- **Languages:** Python, JavaScript, Java, SQL (MySQL, PostgreSQL)
- **Gen AI & LLM:** LLM Fine-tuning, RAG, Prompt Engineering, Agentic AI, Multi-Agent Systems, MCP (Model Context Protocol), AWS Bedrock / AgentCore, OpenAI API (GPT-4o, o1), Claude Sonnet, Llama 2/3
- **Frameworks & Orchestration:** LangChain, LangGraph, Hugging Face Transformers, Next.js, FastAPI
- **Vector Databases & Embeddings:** Titan Embeddings, Pinecone, ChromaDB, pgvector, Semantic Search
- **MLOps & Cloud:** AWS (Bedrock, S3, Lambda), Azure AI / Azure OpenAI, Docker, Git, GitHub, Azure DevOps
- **Tools:** Cursor, GitHub Copilot, LangSmith, Jupyter, Postman

EXPERIENCE

Cognizant Technology Solutions Corporation

Senior Associate – Gen AI Tech Lead

GIFT City, Gandhinagar, Gujarat

May 2025 – Present

► **Project:** *AIG Data Sync PCS – Agentic Framework for Automated Data Pipeline Configuration*

- **Agentic AI Framework:** Architected a generic, reusable **agentic AI framework** on **AWS Bedrock/AgentCore** to automate migration of legacy data-pipeline configs (**IICS XML**, **DQ Validations**, **Transformations**) into standardized **Atlantic YAML**, eliminating manual reverse-engineering effort for a large-scale **GDW modernization** initiative.
- **Multi-Agent Pipeline:** Engineered a **GenAI-driven multi-agent pipeline** — parsing, generation, and validation agents — powered by **Claude Sonnet**, reducing config-migration effort by cutting manual translation steps across complex ETL transformation mappings.
- **RAG & Vector Retrieval:** Built a **RAG-based knowledge retrieval system** (AWS Titan embeddings + vector store) grounding all LLM-generated configs in existing enterprise mapping conventions, significantly improving **output accuracy and consistency** over zero-shot generation.
- **Architecture & PoC Delivery:** Delivered end-to-end **architecture design**, **AWS infrastructure blueprint**, and a working **PoC** validating automated config generation from legacy templates — demonstrating production viability to enterprise stakeholders.

OpsHub Technologies Pvt. Ltd.

Principal Software Engineer

Ahmedabad, Gujarat

June 2020 – April 2025

► **Project:** *OpsHub Data Bridge – LLM-Powered Knowledge & Analytics Hub*

- **LLM Knowledge Hub:** Led team design and deployment of a production **RAG-based knowledge hub** powered by **GPT-3.5 / GPT-4o** via OpenAI APIs and **LangChain**, delivering real-time context-aware insights across **5+ business units** at **100K+ queries/day**.
- **Requirement Analyzer:** Architected an **AI-powered Requirement Analyzer** ingesting tickets from Jira, Azure DevOps, and GitHub to proactively surface planning bottlenecks — reducing project delays by **25%**.
- **MCP Server Development:** Designed and shipped **MCP (Model Context Protocol) servers** enabling secure, context-aware, low-latency integrations between AI agents, enterprise tools, and backend systems.

► **Project:** *OpsHub Insights – Quality Intelligence Engine*

- **Platform Architecture:** Pioneer developer and tech lead for a **unified engineering analytics platform** ingesting raw data from GitHub, Jira, and CI/CD pipelines into real-time, actionable dashboards; architected scalable **multi-source ALM/SCM data pipelines** with reverse ETL for seamless delivery.
- **AI-Assisted Features:** Delivered **12+ AI-assisted analytical features** using domain-specific heuristics and ML models, improving **change impact detection** and **commit analysis accuracy by 30%**, directly enhancing release decision-making efficiency.
- **Performance Engineering:** Achieved **50% system efficiency gain** via multi-layer SQL query optimization and data-layer tuning on **10M+ record datasets**; built a **rule-based NLP approval framework** to automate code reviews and enforce release quality gates.

Accenture India Pvt. Ltd.
Application Development Associate

Pune, Maharashtra
July 2017 – April 2018

► **Project:** *Salesforce CRM Implementation – Comcast Corporation*

- **Salesforce Development:** Developed **Apex triggers and Lightning Web Components** that cut manual data entry by **60%** and measurably improved UI responsiveness and user experience for enterprise CRM workflows.
- **CI/CD & Test Automation:** Shipped integrated Salesforce modules with Jenkins pipelines and automated testing, improving deployment reliability and reducing bug turnaround time.

EDUCATION

Dhirubhai Ambani Institute of Information and Communication Technology
M.Tech – ICT with AI/ML specialization; CPI: 9.60 — President’s Gold Medal

Gandhinagar, Gujarat
2018 – 2020

L.D. College of Engineering, Ahmedabad
B.E. – Computer Engineering; CPI: 9.10

Ahmedabad, Gujarat
2013 – 2017

CERTIFICATIONS

- **AWS Certified AI Practitioner:** Amazon Web Services (AWS) – 2026
- **Claude Certified Architect - Professional:** Anthropic – 2026
- **Claude Certified Architect - Foundations:** Anthropic – 2026
- **Microsoft Certified: Azure Data Scientist Associate:** Microsoft – 2023

ACHIEVEMENTS

- **Men’s Singles & Doubles Table Tennis Champion:** Annual Corporate Sports Tournament — skill, teamwork, and competitive excellence. **(July 2025)**
- **President’s Gold Medal:** Awarded at DA-IICT for highest academic performance across the entire M.Tech program. **(June 2020)**
- **2nd Position – Kaizen Final Year Project Competition:** L.D. College of Engineering, Ahmedabad. **(May 2017)**
- **Top 25 Coders – GTU Coding Competition:** Ranked among the top 25 coders at Gujarat Technological University. **(October 2015)**
- **3rd Position – Zone-Level Chess Competition:** Zone-level scholastic Chess Championship. **(2009)**